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EDITORIAL



Applying click chemistry principles to the design of tumor-targeted nanosystems

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For all advances in creating potent anti-cancer agents, their success is consistently undermined by a fundamental, and frankly, humbling challenge: targeted delivery. Systemic administration inevitably leads to off-target toxicity, limiting safe dosage and harming healthy tissues. Furthermore, tumors erect formidable physical defenses; their dense microenvironment and chaotic vasculature create high internal pressure that blocks drug penetration. Specialized anatomical fortresses, most notably the blood–brain barrier, render entire organs almost inaccessible [1,2]. This failure to accumulate therapeutic concentrations precisely at the site of disease is not a minor hurdle – it is the central bottleneck that stifles innovation and a primary reason why many of the most promising treatments fall short in the clinic.

For decades, the promise of cancer nanomedicine has been to deliver therapies directly to tumors, sparing healthy tissue. While researchers have long focused on overcoming biological hurdles like the Enhanced Permeability and Retention (EPR) effect, stromal density and complex nano-bio interfaces, an equally significant challenge has been the chemistry itself with inherent limitations on the architectural complexity and homogeneity of our designs [3–5].

The advent of click chemistry, pioneered by K. Barry Sharpless, Morten Meldal, and Carolyn Bertozzi, provided a revolutionary solution. At its core, click chemistry is like molecular LEGOs: a set of highly reliable, specific, and efficient reactions that allow scientists to “click” different molecular components together. This approach has transformed nanoparticle design from an art of approximation to a science of precision [6,7]. This synthetic concept is not merely an incremental improvement; it is the arrival of a new chemical paradigm that enables precise molecular engineering and rational assembly of molecularly defined therapeutic systems. The principles of this chemistry, namely modularity, quantitative yields, chemoselectivity, and bioorthogonality, moving us beyond the statistical uncertainties of older conjugation methods [8,9]. Although several reactions meet the requirements, the 1,3-dipolar cycloaddition of terminal alkynes and azides to 1,2,3-triazole has been the most popular.

This development has facilitated the assembly of multifunctional nanosystems with the fidelity of a molecular architect, thereby transforming concepts that were once considered

beyond the limits of what was possible into tangible, reproducible, and highly controlled realities. This transition from a process of functionalisation that occurred incidentally to one of the assembly that is deliberate and designed has, at last, established the robust chemical basis necessary to achieve the full potential of targeted nanomedicine in clinical practice.

This new level of control is already making a profound impact. We can now construct sophisticated nanosystems with unprecedented accuracy [8,9]. Active targeting ligands, including peptides for brain delivery and complex monoclonal antibodies, can now be attached site-specifically to, for instance, lipid nanoparticle surfaces. This ensures optimal orientation and preserves biological function, facilitating systemic mRNA delivery across the blood–brain barrier [10]. Equally significant is its role in enabling the design of biomimetic extracellular vesicles, as illustrated by the process underlying their production. The mimetics are first enriched with bioactive ions, then finished with bone-targeting alendronate ligands, added using metabolic glycoengineering and copper-free Strain-Promoted Alkyne-Azide Cycloadditions (SPAAC) ligation [11]. In each case, click chemistry ensures a uniform product with controlled ligand stoichiometry and orientation. This process transforms the nanoparticle from a mixture of elements into a true molecular machine. It is noteworthy that this novel paradigm of precision extends beyond mere spatial control to encompass kinetic strategy. As the field matures, it is becoming increasingly recognized that the ‘best’ click reaction is not necessarily the ‘fastest.’ Indeed, the rational choice of ligation kinetics is itself a critical design parameter [12]. A slower, more deliberate reaction might be optimal for generating exceptionally stable products, such as an imaging probe designed for long-term tracking. Conversely, the rapid kinetics of a different reaction class becomes paramount when dealing with unstable biomolecules or time-sensitive processes. This enhanced understanding has a significant impact on the selection process, as it provides a substantially expanded and synthetically accessible toolbox. Consequently, we are now in a position to select the optimal reaction rate for a given biological context [8,9,12].

It is important to note that this approach to precision extends beyond the field of therapeutic delivery, reaching the multimodal theranostic and analytical tools that are defining the leading edge of our field. The principle of

orthogonality, defined as the capacity to execute multiple, mutually exclusive reactions within a single vessel, has enabled the development of complex platforms [13]. Recent studies on iridium(III) polypyridine complexes illustrate this, where these luminophores are used as versatile click-ready probes for multimodal bioimaging [14]. Beyond this, we must also consider that a nanoparticle platform can be precisely built through sequential, orthogonal click reactions, layering distinct functionalities. This design enables correlative analyses that combine time-resolved photoluminescence lifetime imaging with high-resolution elemental mapping using synchrotron-based X-ray fluorescence nanoimaging. Coupled with bioorthogonal chemistry to detect metabolic reporters in cells, these integrated datasets provide unprecedented molecular and elemental insight. This approach exemplifies the seamless fusion of synthetic chemistry and advanced imaging, where the chemistry that constructs nanoparticles concurrently illuminates their dynamic biological interactions [15]. Of particular note is that click chemistry's modular precision is revolutionizing biomolecular analysis by enabling the selective enrichment of rare targets – such as phosphoinositides – via “click cluster” constructs on magnetic beads. This powerful approach streamlines advanced mass spectrometry and brings a remarkable symmetry to the field: the very chemical principles enabling the assembly of sophisticated nanomedicines are also harnessed for their interrogation [15].

Perhaps the most exciting frontier is using click chemistry inside the body. Instead of delivering a complete, complex drug, we can send in smaller, separate components that find the tumor and then “click” together to form the active therapeutic on-site. This approach, known as pretargeting or *in situ* self-assembly, promises unprecedented specificity and could dramatically reduce side effects [13,16]. Liquid biopsy approach for cancer early detection has been achieved by implementing click chemistry for selective enrichment of cancer-derived extracellular vesicles (EV), enabling sensitive and specific quantification of tumor-associated EV subpopulations from plasma samples [17]. On another front, the extraordinary kinetics of the inverse electron-demand Diels-Alder (IEDDA) reaction are redefining what's possible in clinical imaging and therapy. Pretargeting strategies, now clinically viable, exploit this chemistry with precision: a slow-clearing antibody, equipped with a trans-cyclooctene handle, accumulates at the tumor before a fast-clearing tetrazine-bearing radiopharmaceutical is introduced. The resulting *in vivo* ligation – swift and bioorthogonal – yields high-contrast images and exceptional therapeutic specificity, minimizing radiation exposure to healthy tissue while targeting malignancy with remarkable accuracy. This elegant interplay of timing and chemistry is setting new standards in precision cancer care [18]. Recent studies have demonstrated that large, complex PROTACs (proteolysis-targeting chimeras), which often suffer from poor pharmacokinetic properties, can be deconstructed into two smaller, drug-like precursors [19]. These components, tagged with complementary bioorthogonal handles, can be co-encapsulated into a single targeted nanocarrier. Upon delivery to the tumor, they are released and subsequently self-assemble via an IEDDA reaction, forming the active PROTAC precisely at its site of action [20]. This strategy represents

a fundamental shift from simply delivering a drug to delivering the instructions for its final synthesis exactly where it is needed.

Successful application requires careful attention to the *in vivo* stability of click-derived linkers (eg hydrolysis, enzymatic degradation, and competitive exchange) to prevent premature release or detachment of targeting moieties – which would compromise tumor selectivity and reduce therapeutic efficacy. Therefore, potential immunogenicity from newly introduced chemical motifs or residual catalysts has to be considered. Then, the complexities of maintaining reproducibility and purity during large-scale synthesis are an additional key point. Addressing these challenges is essential to translate click-modified nanosystems for clinical applications, balancing advanced functionality with safety, manufacturability, and regulatory compliance [2–5,21].

The evolution of click chemistry continues to open new doors for nanomedicine. Beyond the established reactions, a powerful new suite of tools – including Sulfur fluoride exchange (SuFEx), PFEx, and SeNEx – is emerging [8]. These chemistries introduce unique reactive properties that are highly advantageous for building therapeutic nanosystems. The exceptional stability of SuFEx linkages provides the durability needed for *in vivo* applications, while PFEx and SeNEx expand our ability to selectively modify complex biological targets. Perhaps most importantly, their metal-free character removes a significant hurdle for clinical translation by reducing potential toxicity. In embracing these new reactions, the field is not just adding more tools, but enhancing its capacity for the modular, efficient, and reproducible synthesis of precisely targeted, next-generation nanotherapeutics. Ultimately, click chemistry provides the robust and versatile foundation that nanomedicine has long needed. It has shifted our focus from simply hoping a nanoparticle reaches its target to rationally designing it to succeed. By providing the tools for precise molecular engineering, this chemical paradigm is not just improving our existing strategies – it is enabling us to build the next generation of truly targeted cancer therapies and diagnostics. The future of nanomedicine is not just being discovered; it is being deliberately and precisely constructed, one click at a time.

Author contributions

RMSM conceived the idea for the editorial, drafted the initial manuscript, coordinated the revision process, revised the final version of the document and assisted in structuring the text. JJDM provided conceptual input, contributed key references, critically revised the content, and reviewed and edited the initial version of the manuscript for submission.

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References

1. Zafar A, Khatoon S, Khan MJ, et al. Advancements and limitations in traditional anti-cancer therapies: a comprehensive review of surgery, chemotherapy, radiation therapy, and hormonal therapy. *Discov Oncol.* **2025**;16(1):607. doi: [10.1007/s12672-025-02198-8](https://doi.org/10.1007/s12672-025-02198-8)
2. Ortega-Liebana MC, Sanchez-Martin RM. Nanomedicine basics. *ACS In Focus.* **2025**. doi: [10.1021/acsinfocus.7e9013](https://doi.org/10.1021/acsinfocus.7e9013)
3. Mitchell M, Billingsley M, Haley R, et al. Engineering precision nanoparticles for drug delivery. *Nat Rev Drug Discov.* **2021**;20:101–124. doi: [10.1038/s41573-020-0090-8](https://doi.org/10.1038/s41573-020-0090-8)
4. Anchordoquy T, Artzi N, Balyasnikova I, et al. Mechanisms and barriers in nanomedicine: progress in the field and future directions. *ACS Nano.* **2024**;18(22):13983–13999. doi: [10.1021/acs.nano.4c00182](https://doi.org/10.1021/acs.nano.4c00182)
5. Tiwari R, Kolli M, Chauhan S, et al. Tabletized nanomedicine: from the current scenario to developing future medicine. *ACS Nano.* **2024**;18(18):11503–11524. doi: [10.1021/acsnano.4c00014](https://doi.org/10.1021/acsnano.4c00014)
6. Devaraj NK, Finn MG. Introduction: click chemistry. *Chem Rev.* **2021**;121(12):6697–6698. doi: [10.1021/acs.chemrev.1c00469](https://doi.org/10.1021/acs.chemrev.1c00469)
7. Wu P. The Nobel Prize in Chemistry 2022: fulfilling demanding applications with simple reactions. *ACS Chem Biol.* **2022**;17(11):2959–2961. doi: [10.1021/acscembio.2c00788](https://doi.org/10.1021/acscembio.2c00788)
8. Ghosal K, Bhattacharyya SK, Mishra V, et al. Click chemistry for biofunctional polymers: from observing to steering cell behavior. *Chem Rev.* **2024**;124(23):13216–13300. doi: [10.1021/acs.chemrev.4c00251](https://doi.org/10.1021/acs.chemrev.4c00251)
9. Mehak SG, Singh R, Singh R, et al. Clicking in harmony: exploring the bio-orthogonal overlap in click chemistry. *RSC Adv.* **2024**;14(11):7383–7413. doi: [10.1039/D4RA00494A](https://doi.org/10.1039/D4RA00494A)
10. Han EL, Tang S, Kim D, et al. Peptide-functionalized lipid nanoparticles for targeted systemic mRNA delivery to the brain. *Nano Lett.* **2025**;25(2):800–810. doi: [10.1021/acs.nanolett.4c05186](https://doi.org/10.1021/acs.nanolett.4c05186)
11. Jiang H, Zhu X, Yu J, et al. Biomimetic extracellular vesicles based on composite bioactive ions for the treatment of ischemic bone disease. *ACS Nano.* **2024**;18(51):34924–34948. doi: [10.1021/acs.nano.4c13028](https://doi.org/10.1021/acs.nano.4c13028)
12. Luu T, Gristwood K, Knight JC, et al. Click chemistry: reaction rates and their suitability for biomedical applications. *Bioconjugate Chem.* **2024**;35(6):715–731. doi: [10.1021/acs.bioconjchem.4c00084](https://doi.org/10.1021/acs.bioconjchem.4c00084)
13. Khandelwal D, Bhattacharya A, Kumari V, et al. Leveraging nano-materials for ultrasensitive biosensors in early cancer detection: a review. *J Mater Chem B.* **2025**;13(3):802–820. doi: [10.1039/d4tb02107j](https://doi.org/10.1039/d4tb02107j)
14. Rigolot V, Simon C, Bouchet A, et al. Click-ready iridium(III) complexes as versatile bioimaging probes for bioorthogonal metabolic labeling. *RSC Chem Biol.* **2025**;6(3):364–375. doi: [10.1039/d4cb00255e](https://doi.org/10.1039/d4cb00255e)
15. Nguyen DPL, Le HT, Kim DH, et al. Enrichment and MALDI-TOF-MS/MS analysis of phosphatidylinositol bisphosphates in brain tissue. *Anal Biochem.* **2025**;698:115749. doi: [10.1016/j.ab.2024.115749](https://doi.org/10.1016/j.ab.2024.115749)
16. Bauer D, Cornejo MA, Hoang T, et al. Click chemistry and radiochemistry: an update. *Bioconjugate Chem.* **2023**;34(11):1925–1950. doi: [10.1021/acs.bioconjchem.3c00286](https://doi.org/10.1021/acs.bioconjchem.3c00286)
17. Zhao C, Wang Z, Kim H, et al. Identification of tumor-specific surface proteins enables quantification of extracellular vesicle subtypes for early detection of pancreatic ductal adenocarcinoma. *Adv Sci.* **2025**;12(21):2414982. doi: [10.1002/adv.202414982](https://doi.org/10.1002/adv.202414982)
18. Yuen D, Feeney OM, Noi L, et al. Nanobody-mediated cellular uptake maximizes the potency of polylysine dendrimers while preserving solid tumor penetration. *ACS Nano.* **2025**;19(6):6044–6057. doi: [10.1021/acsnano.4c10851](https://doi.org/10.1021/acsnano.4c10851)
19. Yang C, Tripathia R, Wang B. Click chemistry in the development of PROTACs. *RSC Chem Biol.* **2024**;5(3):189–197. doi: [10.1039/D3CB00199G](https://doi.org/10.1039/D3CB00199G)
20. Xie S, Zhu J, Peng Y, et al. In vivo self-assembly of PROTACs by bioorthogonal chemistry for precision cancer therapy. *Angew Chem Int Ed.* **2025**;64(11):e202421713. doi: [10.1002/anie.202421713](https://doi.org/10.1002/anie.202421713)
21. Baek M-J, Hur W, Kashiwagi S, et al. Design considerations for organ-selective nanoparticles. *ACS Nano.* **2025**;15(15):14605–14626. doi: [10.1021/acsnano.5c00484](https://doi.org/10.1021/acsnano.5c00484)